

Definition 11.1

Let $\alpha \in S_n$ and let $i \in \{1, \dots, n\}$. We say that α *moves* i if $\alpha(i) \neq i$. If $\alpha(i) = i$ we say that α *fixes* i .

Example. The permutation

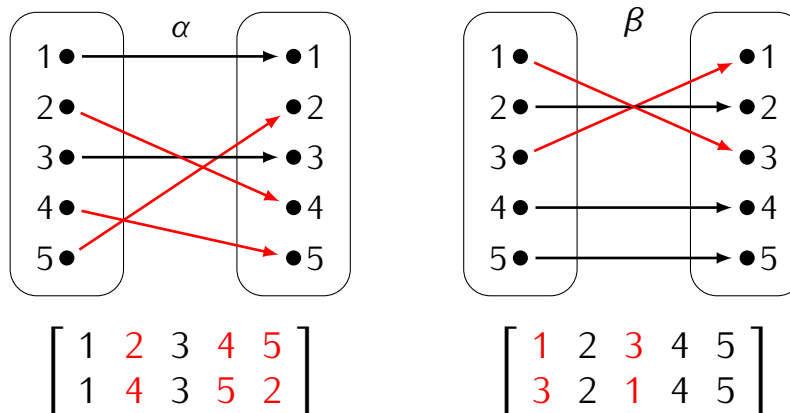
$$\begin{bmatrix} 1 & 2 & 3 & 4 \\ 2 & 4 & 3 & 1 \end{bmatrix}$$

moves 1, 2 and 4, and fixes 3.

Definition 11.2

We will say that permutations $\alpha, \beta \in S_n$ are *disjoint* if there is no $i \in \{1, \dots, n\}$ which is moved by both α and β .

Example.

**Theorem 11.3**

If $\alpha, \beta \in S_n$ are disjoint permutations then $\alpha \circ \beta = \beta \circ \alpha$. Moreover,

$$\alpha \circ \beta(i) = \begin{cases} \alpha(i) & \text{if } i \text{ is moved by } \alpha \\ \beta(i) & \text{if } i \text{ is moved by } \beta \\ i & \text{otherwise} \end{cases}$$

Proof. Assume that $i \in \{1, \dots, n\}$ is an element moved by α . Then α also moves $\alpha(i)$. It follows that both i and $\alpha(i)$ are fixed by β , so we have

$$\beta \circ \alpha(i) = \alpha(i) = \alpha \circ \beta(i)$$

By the same argument, if i is moved by β then

$$\alpha \circ \beta(i) = \beta(i) = \beta \circ \alpha(i)$$

Finally, if both α and β fix i then

$$\alpha \circ \beta(i) = i = \beta \circ \alpha(i)$$

□

Definition 11.4

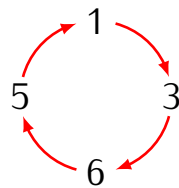
A permutation $\alpha \in S_n$ is a *cycle of length r* (or *r -cycle*) if there are distinct elements $i_1, i_2, \dots, i_r \in \{1, 2, \dots, n\}$ such that

$$\alpha(i_1) = i_2, \quad \alpha(i_2) = i_3, \quad \dots \quad \alpha(i_{r-1}) = i_r, \quad \alpha(i_r) = i_1$$

and α fixes all other elements of $\{1, \dots, n\}$.

Example.

$$\alpha = \begin{bmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 3 & 2 & 6 & 4 & 1 & 5 \end{bmatrix}$$



Cycle notation. A permutation α such that

$$\alpha(i_1) = i_2, \quad \alpha(i_2) = i_3, \quad \dots \quad \alpha(i_{r-1}) = i_r, \quad \alpha(i_r) = i_1$$

and which fixes all other elements is denoted by $(i_1, i_2, \dots, i_r)$.

Example.

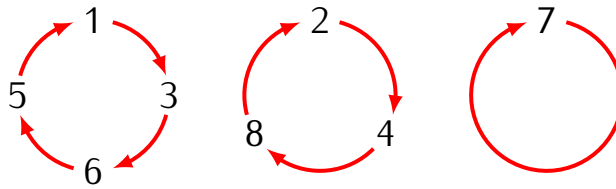
$$\begin{bmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 3 & 2 & 5 & 4 & 6 & 1 \end{bmatrix} = (1, 3, 5, 6) = (3, 5, 6, 1) = (5, 6, 1, 3) = (6, 1, 3, 5)$$

Theorem 11.5

Every permutation in S_n is either a cycle or a product of disjoint cycles.

Example.

$$\begin{bmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 \\ 3 & 8 & 6 & 2 & 1 & 5 & 7 & 4 \end{bmatrix} = (1, 3, 6, 5) \circ (2, 8, 4) \circ (7) = (1, 3, 6, 5) \circ (2, 8, 4)$$



Lemma 11.6

Let $\alpha \in S_n$, and let $i_0 \in \{1, \dots, n\}$ be an element moved by α . Then:

- 1) There exists $r > 1$ such that $\alpha^r(i_0) = i_0$
- 2) If $r > 1$ is the smallest integer satisfying $\alpha^r(i_0) = i_0$ then all elements

$$i_0, \alpha(i_0), \alpha^2(i_0), \dots, \alpha^{r-1}(i_0)$$

are distinct.

Proof. Consider the sequence

$$i_0 = \alpha^0(i_0), \alpha^1(i_0), \alpha^2(i_0), \dots$$

Since all elements of this sequence come from the finite set $\{1, \dots, n\}$, there must be an integer $r \geq 1$ such that the elements $i_0, \alpha(i_0), \alpha^2(i_0), \dots, \alpha^{r-1}(i_0)$ are distinct and $\alpha^r(i_0)$ is equal to one of the previous elements. We will show that $\alpha^r(i_0) = i_0$. Indeed, otherwise $\alpha^r(i_0) = \alpha^k(i_0)$ for some $1 \leq k < r$. This gives

$$\alpha(\alpha^{r-1}(i_0)) = \alpha(\alpha^{k-1}(i_0))$$

and since α is a 1-1 function, we obtain

$$\alpha^{r-1}(i_0) = \alpha^{k-1}(i_0)$$

This contradicts the assumption that the elements $i_0, \alpha(i_0), \dots, \alpha^{r-1}(i_0)$ are distinct. $\square$

Proof of Theorem 11.5. Let $\alpha \in S_n$. We will argue that α can be written as a product of cycles by induction with respect to the number k of elements of $\{1, \dots, n\}$ moved by α . If $k = 0$ then α fixes all elements, so it is the identity permutation, which is a 1-cycle.

Assume then that all permutations moving k or fewer elements can be written as a product of disjoint cycles and that α moves $k + 1$ elements. Let $i_0 \in \{1, \dots, n\}$ be an element moved by α . By Lemma 11.6 there is $r > 1$ such that the elements

$$i_0, \alpha(i_0), \alpha^2(i_0), \dots, \alpha^{r-1}(i_0)$$

are all distinct and $\alpha^r(i_0) = i_0$. Denote for $k = 1, \dots, r-1$ denote $i_k = \alpha^k(i_0)$. Notice that $\alpha(i_k) = i_{k+1}$ for $k < r-1$ and $\alpha(i_{r-1}) = i_0$. Let $\beta \in S_n$ be a permutation defined as follows:

$$\beta(i) = \begin{cases} i & \text{if } i \in \{i_0, \dots, i_{r-1}\} \\ \alpha(i) & \text{otherwise} \end{cases}$$

Notice that the cycle $(i_0, i_1, \dots, i_{r-1})$ and β are disjoint permutations. Thus, we can use Theorem 11.3 to show that $\alpha = (i_0, i_1, \dots, i_{r-1}) \circ \beta$. Then, since β moves fewer elements than α , by the inductive assumption we can write β as a product of disjoint cycles:

$$\beta = \gamma_1 \circ \dots \circ \gamma_m$$

Therefore we obtain a decomposition of α into a product of disjoint cycles:

$$\alpha = (i_0, i_1, \dots, i_{r-1}) \circ \gamma_1 \circ \dots \circ \gamma_m$$

$\square$

Recall that the least common multiple of integers $n_1, n_2, \dots, n_k \geq 1$ is the smallest positive integer $\text{lcm}(n_1, \dots, n_k)$ which is divisible by each of these integers.

Theorem 11.7

Assume that a permutation $\alpha \in S_n$ has a decomposition into disjoint cycles

$$\alpha = \gamma_1 \circ \cdots \circ \gamma_m$$

where γ_i is a cycle of length $r_i > 1$. Then the order of α is given by

$$|\alpha| = \text{lcm}(r_1, r_2, \dots, r_m)$$

Proof. First, notice that if γ is an r -cycle then $|\gamma| = r$. Let

$$\alpha = \gamma_1 \circ \cdots \circ \gamma_m$$

where γ_i is an r_i -cycle, and let $p = \text{lcm}(r_1, \dots, r_m)$. By Theorem 11.3 disjoint cycles commute, so

$$\alpha^p = (\gamma_1 \circ \cdots \circ \gamma_m)^p = \gamma_1^p \circ \cdots \circ \gamma_m^p = \varepsilon$$

where ε is the identity permutation. By Theorem 6.3 we obtain that $|\alpha|$ divides p .

Next, we claim that $\gamma_i^{|\alpha|} = \varepsilon$ for each i . Indeed, since the cycles are disjoint, elements moved by $\gamma_i^{|\alpha|}$ are fixed by $\gamma_j^{|\alpha|}$ for all $j \neq i$, so if $\gamma_i^{|\alpha|}$ moves some element, then the same element is moved by $\gamma_1^{|\alpha|} \circ \cdots \circ \gamma_m^{|\alpha|}$. This however cannot happen because

$$\gamma_1^{|\alpha|} \circ \cdots \circ \gamma_m^{|\alpha|} = (\gamma_1 \circ \cdots \circ \gamma_m)^{|\alpha|} = \alpha^{|\alpha|} = \varepsilon$$

In this way we obtained that r_i divides $|\alpha|$ for $i = 1, \dots, m$, and so p divides $|\alpha|$. Therefore $|\alpha| = p$.

□

Exercise. Compute the order of the following permutation in S_8 :

$$\alpha = \begin{bmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 \\ 3 & 8 & 6 & 2 & 1 & 5 & 7 & 4 \end{bmatrix}$$

Exercise. Find all possible orders of elements of S_5 .

Exercise. Compute the number of permutations of order 10 in S_8 .

Exercise. Compute the number of permutations of order 3 in S_7 .